



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-12

Tic-Tac-Toe Game using Python

Aim

To write a Python program to implement the Tic-Tac-Toe Game.

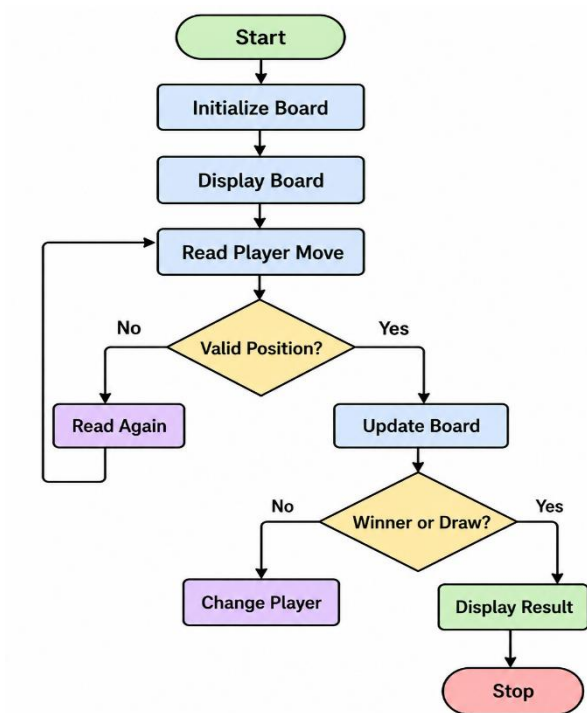
Objective

To implement a two-player Tic-Tac-Toe game.

Algorithm

1. Start.
2. Initialize a 3×3 game board.
3. Display the board.
4. Read the row and column from the current player.
5. Place the player's symbol (X or O) if the position is empty.
6. Check for a winner or draw.
7. If the game is not over, switch the player and repeat.
8. Display the result and stop.

Flowchart



Code

```
board = [" "] * 9

def show():
    for i in range(0, 9, 3):
        print(board[i], "|", board[i+1], "|", board[i+2])
    if i < 6:
        print("--+---+--")

def win(p):
    w = [(0,1,2),(3,4,5),(6,7,8),
          (0,3,6),(1,4,7),(2,5,8),
          (0,4,8),(2,4,6)]
    return any(board[a]==board[b]==board[c]==p for a,b,c in w)
```

```
player = "X"
```

```
for _ in range(9):
```

```
    show()
```

```
    pos = int(input(f"Player {player}, enter position (1-9): ")) - 1
```

```
    if board[pos] != " ":
```

```
        print("Position already taken!")
```

```
        continue
```

```
    board[pos] = player
```

```
    if win(player):
```

```
        show()
```

```
        print(f"Player {player} Wins!")
```

```
        break
```

```
    player = "O" if player == "X" else "X"
```

```
else:
```

```
    show()
```

```
    print("Game Draw!")
```

Output



The screenshot shows a VS Code editor with a Python script for a Tic Tac Toe game. The script is in a file named 'exp 12.py'. The code defines a 3x3 board, a show function, a win function, and a main loop. The output in the terminal shows the game progress:

```
Player O, enter position (1-9): 8
X | X | 
-----
| O | 
-----
| O | 
-----
Player X, enter position (1-9): 3
X | X | X
-----
| O | 
-----
| O | 
-----
Player X Wins!
```

Result

The Tic-Tac-Toe game was executed successfully, and the winner (or draw) was displayed correctly.

Conclusion

The Tic-Tac-Toe game was implemented successfully using Python with two-player interaction and win/draw detection.